Experiment-15

Write a python program to implement Decision Tree

Aim

To write and execute a Python program to build a Decision Tree using the ID3 algorithm, where Information Gain is used to select the best attribute, and to predict the class of a new sample.

Algorithm

1. Take the training data with the attributes Outlook, Humidity and Wind and the class Play.
2. Calculate the entropy of the data set.
3. Calculate the Information Gain of each attribute.
4. Select the attribute with the highest Information Gain as the decision node.
5. Split the data set based on the values of the selected attribute.
6. Repeat the process for each subset using the remaining attributes.
7. Stop when all the examples in a subset belong to the same class and make it a leaf node.
8. Display the decision tree.
9. Use the tree to predict the class of a new sample.

Coding

ex-15.py - D:/ex-15.py (3.10.0)

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```
# Decision Tree (ID3 Algorithm)
import math

# Weather data : Outlook, Humidity, Wind -> Play
attributes = ["Outlook", "Humidity", "Wind"]
data = [
    ["Sunny", "High", "Weak", "No"],
    ["Sunny", "High", "Strong", "No"],
    ["Overcast", "High", "Weak", "Yes"],
    ["Rain", "High", "Weak", "Yes"],
    ["Rain", "Normal", "Weak", "Yes"],
    ["Rain", "Normal", "Strong", "No"],
    ["Overcast", "Normal", "Strong", "Yes"],
    ["Sunny", "High", "Weak", "No"],
    ["Sunny", "Normal", "Weak", "Yes"],
    ["Rain", "Normal", "Weak", "Yes"],
    ["Sunny", "Normal", "Strong", "Yes"],
    ["Overcast", "High", "Strong", "Yes"],
    ["Overcast", "Normal", "Weak", "Yes"],
    ["Rain", "High", "Strong", "No"]
]

def entropy(rows):
    total = len(rows)
    result = 0
    for label in set(r[-1] for r in rows):
        p = [r[-1] for r in rows].count(label) / total
        result = result - p * math.log2(p)
    return result
```

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```
def gain(rows, col):
    result = entropy(rows)
    for value in set(r[col] for r in rows):
        subset = [r for r in rows if r[col] == value]
        result = result - len(subset) / len(rows) * entropy(subset)
    return result

def build(rows, cols):
    labels = [r[-1] for r in rows]
    if len(set(labels)) == 1:
        return labels[0]
    if not cols:
        return max(set(labels), key=labels.count)

    best = max(cols, key=lambda c: gain(rows, c))
    tree = {attributes[best]: {}}
    rest = [c for c in cols if c != best]
    for value in set(r[best] for r in rows):
        subset = [r for r in rows if r[best] == value]
        tree[attributes[best]][value] = build(subset, rest)
    return tree

def show(tree, indent=0):
    for attr, branches in tree.items():
        for value, sub in sorted(branches.items()):
            print("    " * indent + attr + " = " + value)
            if isinstance(sub, dict):
                show(sub, indent + 1)
            else:
                print("    " * (indent + 1) + "-> Play : " + sub)
```

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```
print("DECISION TREE (ID3)")
print("--------------------")
for i in range(3):
    print("Information Gain of", attributes[i], "=",
          round(gain(data, i), 3))
print()
tree = build(data, [0, 1, 2])
show(tree)

# Predict for a new sample
print()
sample = {"Outlook": "Rain", "Humidity": "High", "Wind": "Weak"}
print("New sample :", sample)
node = tree
while isinstance(node, dict):
    attr = list(node.keys())[0]
    node = node[attr][sample[attr]]
print("Prediction : Play =", node)
```

Output

IDLE Shell 3.10.0

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```
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========================== RESTART: D:/ex-15.py ==========================
DECISION TREE (ID3)
--------------------
Information Gain of Outlook = 0.247
Information Gain of Humidity = 0.152
Information Gain of Wind = 0.048

Outlook = Overcast
    -> Play : Yes
Outlook = Rain
    Wind = Strong
        -> Play : No
    Wind = Weak
        -> Play : Yes
Outlook = Sunny
    Humidity = High
        -> Play : No
    Humidity = Normal
        -> Play : Yes

New sample : {'Outlook': 'Rain', 'Humidity': 'High', 'Wind': 'Weak'}
Prediction : Play = Yes
>>>
```

Result

The Decision Tree was successfully built using the ID3 algorithm. Outlook has the highest Information Gain (0.247) and is selected as the root node. For the new sample (Rain, High, Weak), the predicted result is Play = Yes.